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Probabilistic Modeling of Thermal Grids using Gaussian Processes

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Abstract— **Dynamic physics based modeling of district heating networks has gained importance due to an increased use of renewable energy sources and a transition towards lower temperature district heating networks. The modeling is enhanced by technologies for automatic model generation and co-simulation. These models are in general not suitable for automatic control and optimization methods, due to the complexity of the model. Moreover, there is no notion of uncertainty in the models, something that can be of importance for decision making, and that can be explicitly accounted for in e.g Bayesian Optimization and Stochastic Nonlinear Model Predictive Control. In this paper a data driven Gaussian process model for the thermal dynamics of the district heating grid is proposed, with a kernel derived using known physics and numerical methods. The model is trained and validated on a realistic first principle simulation model of a district heating pipe. Results show a good correspondence with the output from the training model on a validation dataset, providing explicit propagation of the input uncertainties. It is suggested that the method can be scaled up to larger parts of the grid for use in advanced control and optimization methods.**

I. Introduction

Dynamic simulation of district heating and cooling grids is a topic of numerous research projects and papers. Most articles use first principle models using known physical relationships [1], [2], usually in acausal equation based languages such as Modelica, to model the grid. Since the number of measurements are usually scarce in actual district heating and cooling grids, while the physical relationships are well known and validated, these models augment understanding of the process and provide a valuable tool e.g for grid design, and validation of automatic control, scheduling and optimization schemes. Despite the upside to physical modeling, for automatic control and optimization the complexity of the model often makes it unsuitable. There is also no notion of uncertainty in the results, even though the model often relies on uncertain predictions for e.g weather and consumer heat load. The uncertainty can, if accounted for, be used as an aid in decision making, and for control methods such as stochastic nonlinear model predictive control (NMPC) [3].

The aim of the paper is to lay the foundation for probabilistic modeling of district heating networks using Gaussian processes, where a first principle model is used for training and validation of the method. The uncertainty of the predictions are explicitly accounted for by propagation of the variance of the predicted heat loads and outdoor temperature, that are inputs to the model. The knowledge from the physical model is incorporated into the Gaussian process model, inspired by concepts such as scientific machine learning [4] where convolutional artificial neural networks are being used, and numerical Gaussian processes [5] where the structure of a PDE is encoded into the kernel of the Gaussian process. It is shown that carefully tailoring the method to the process can circumvent many of the common drawbacks for data driven methods in general, and Gaussian processes in particular.

The paper is organized as follows. A physical model of a district heating pipe is first presented, including common numerical methods solving the corresponding partial differential equations. Gaussian process regression is then introduced, where the methods and choice of kernels are derived using the physics based model and corresponding numerical methods. The method is then validated against a well verified pipe model with realistic thermal behavior for a district heating network. Finally, conclusions and proposed further research are given.

II. Preliminaries

For this paper use cases with time scales ranging from minutes to hours are considered. The pressure dynamics are on a time scale of seconds and can thus be approximated as static, and are not considered in the paper. The thermal dynamics are several magnitudes slower, and are of large interest when exploring use cases such as utilization of the thermal inertia of the network, and as an aid to understanding the grid dynamics. For the mentioned purposes the spatial thermal distribution needs to be modeled.

For different use cases the uncertainty can serve different purposes. For simulation used to validate functionality, input uncertainty can be seen as a feature, where sampling from a distribution prevents the overly deterministic behavior of using recorded time series or deterministic models. For automatic control and optimization, accounting for the uncertainty can enhance the performance of the methods [6], but also comes with a computational cost.

III. Physical Model of a District Heating Pipe

The main components of a district heating grid are the piping, consumers, producers and pumps. For this paper we limit ourselves to the pipe and consumer part of the network. A schematic view of the process can be seen in Figure 1. For both supply and return pipes, water of varying temperature

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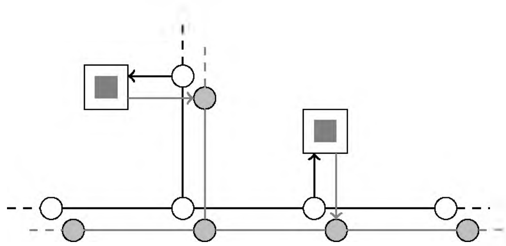

Fig. 1. Consumers (rectangles) draw hot water from the supply pipes (black line), and return colder water to the return pipes (gray line) at specific points, often referred to as nodes (circles).

is injected at different locations, and there are losses along the pipe length. To simulate a scenario the injected water flow and temperature needs to be modeled using predictions or known time series.

The thermal dynamics of a district heating pipe can be described by a 1D partial differential equation (PDE), where axial diffusion, pressure losses, dissipation and wall friction have been shown to have a negligible impact [7] on the temperature for the operational ranges of district heating, and are thus neglected. This gives the equation

$$\rho c_p A \frac{dT(x,t)}{dt} + q(x,t) = -\rho c_p A v(t) \frac{dT(x,t)}{dx}, \quad (1)$$

where T is the temperature, ρ is the density, c_p is the heat capacity, A is the cross sectional area of the pipe, $q(x,t)$ the heat loss to the surroundings and $v(t)$ the flow velocity. Heat losses to surroundings are usually approximated by a function of the temperature difference between the water $T(t)$ and the temperature of the surroundings $T_o(t)$

$$q(x,t) = f_{ht}(T(x,t) - T_o(x,t)). \quad (2)$$

In this article it is assumed that there is perfect mixing for every node, so that the input to each pipe joining two nodes is for k joining flows

$$\dot{m}_{tot} = \sum_{i=1}^{k} \dot{m}_i, \qquad T_{tot} = \frac{1}{\dot{m}_{tot}} \sum_{i=1}^{k} \dot{m}_i T_i \quad (3)$$

where $\dot{m}_i$ are mass flow rates and T_i the corresponding temperatures of each flow. These relations set the boundary conditions for Equation (1) for a pipe joining two nodes. It is further assumed that there is no accumulation of mass in the pipe or node due to the incompressibility of water. The fluid velocity can be calculated, assuming that the cross sectional area A_p is known, as

$$v(t) = \frac{\dot{m}(t)}{\rho A_p}. \quad (4)$$

To solve the PDE (1) numerically the pipe is usually discretized along the pipe length. Applying an upwind discretization scheme [8] we can rewrite (1-2), using subscripts for spatial steps and superscripts for time steps, so that each segment of the pipe can be written as

$$T_i^{k+1} = T_i^k + v^k \frac{\Delta t}{\Delta x}(T_i^k - T_{i-1}^k) - \frac{\Delta t}{\Delta x} f_{ht}(T_i^k - T_{o,i}^k), \quad (5)$$

where v is the flow velocity, Δt the time step, and Δx the spatial step. By introducing the variables

$$\Delta_t T_i^{k+1} = T_i^{k+1} - T_i^k, \qquad \Delta_x T_i^k = T_i^k - T_{i-1}^k \quad (6)$$

$$\Delta T_{o,i}^k = T_i^k - T_o^k \quad (7)$$

equation (5) can be written as

$$\Delta_t T_i^{k+1} = v^k \frac{\Delta t}{\Delta x} \Delta_x T_i^k - \frac{\Delta t}{\Delta x} f_{ht}(\Delta T_{o,i}^k). \quad (8)$$

For the discretization of the grid the Courant-Friedrichs-Lewy (CFL) condition needs to be considered. The CFL condition gives that distance travelled in one timestep must be less than the distance between two points of the grid for stability of an explicit time stepping scheme. This makes intuitive sense, the term T_{i-2}^k will directly impact T_i^{k+1} if more than Δx have been covered. The CFL condition for (most) explicit schemes can be written as

$$\frac{v_{sup} \Delta t}{\Delta x} \leq 1, \quad (9)$$

where the maximum v_{sup} of $v(t)$ is assumed to be known. When $CFL \ll 1$ noteable numerical diffusion will be present. This can be mitigated by a variable step size, but is not considered within the scope of this paper. The equations above will be used when designing the kernels step size for the Gaussian process in the following section. Notably, the above equations are *assumptions* based on known physics and numerical computing, not a carbon copy of the actual model.

IV. Gaussian Process Model

In this section a brief summary of Gaussian process regression for dynamical systems is provided. It is then shown how the known physics can be encoded into the kernel and transform the problem to a computationally efficient data driven surrogate model, where the uncertainty is seamlessly included for multi-step ahead predictions. Flow rates are assumed to be known for the scope of this paper, possibly including uncertainty from predictions.

The Gaussian process regression problem [9] can be summarized as trying to find a distribution over functions

$$f(x) \sim \mathcal{GP}(m(x), k(x, x')) \quad (10)$$

where $m(x)$ is a mean function and $k(x,x')$ is a covariance function for the vectors x, x', that is specified by a chosen kernel. For notational convenience the shorthands $K = k(X, X')$, $k_* = k(X, x_*)$ and $k_{**} = k(x_*, x_*)$ are used, where X is the training dataset, y the target outcomes, x_* a new point at where the prediction is made, and $f_* = f(x_*)$ the prediction at x_*. For this paper a zero mean function will be assumed, further motivated in section IV-A, also providing some notational benefits. For a new point the joint distribution with regards to the already observed points can be written as

$$\begin{pmatrix} y \\ f_* \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} K + \sigma_n I & k_*^\top \\ k_* & k_{**} \end{pmatrix} \right) \quad (11)$$

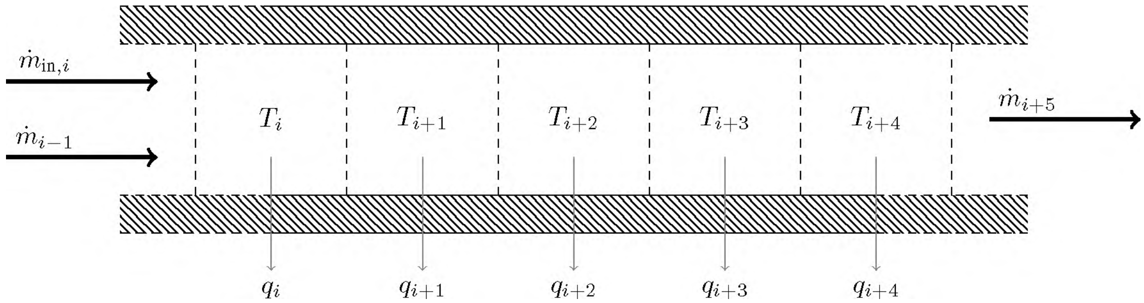

Fig. 2. Overview of a discretized 1D pipe model with finite volumes.

allowing us to analytically derive the predictive distribution for our target, that is

$$f_*|x_*, X, y \sim \mathcal{N}(\bar{f}_*, \text{cov}(f_*)) \tag{12}$$

where the posterior mean and covariance can be calculated in closed form

$$\bar{f}_* = k_*^\top (K + \sigma_n^2 I)^{-1} y \tag{13}$$

$$\text{cov}(f_*) = k_{**} - k_*^\top (K + \sigma_n^2 I)^{-1} k_* \tag{14}$$

Gaussian processes are non-parametric, however the kernel $k(x, x')$ includes hyperparameters that need to be estimated. In this paper the Squared Exponential (SE) kernel is used, given by

$$k(x, x') = \sigma^2 \text{exp}\left(-\frac{(x - x')^2}{2\ell^2}\right), \tag{15}$$

and is parameterized by the length scale ℓ and the output variance σ. The squared exponential (SE) kernel has analytically tractable properties, e.g with regards to multiplication of kernels and differentiation, as will be shown later. The hyperparameters σ and ℓ can be found e.g using the quasi Newton solver L-BFGS, and minimizing the negative log marginal likelihood

$$\log p(y|X) = -\frac{1}{2} y^\top (K + \sigma_n^2 I)^{-1} y \tag{16}$$
$$-\frac{1}{2}\log |K + \sigma_n^2 I| - \frac{\pi}{2}\log 2\pi.$$

A. Gaussian process for a district heating pipe

The discretization of (8) is attractive also from a simulation and visualization standpoint —to augment the understanding of the process, the spatial temperature distribution along the pipe provides valuable information. From the numerical solution (8) a first order Markov chain model is implicitly assumed, that can be written in the form

$$x^{k+1} = x^k + f(x^k, u^k, \epsilon_x^k) \tag{17}$$

$$y^{k+1} = x^{k+1} + \epsilon_y^k \tag{18}$$

$$\epsilon_y^k \sim \mathcal{N}(0, \sigma_y^2) \qquad \epsilon_x^k \sim \mathcal{N}(0, \Sigma_x) \tag{19}$$

forming a so called Gaussian Process State Space Model (GPSSM) when the transition function $f(x^k, u^k, \epsilon_x^k)$ is modeled as a Gaussian process. The noise ϵ_y^k can be interpreted as the numerical uncertainty, and ϵ_x^k is the uncertainty of the inputs and the previous state estimation, that will be propagated through the Gaussian process model. The states are fully observed, so that standard Gaussian process regression [10] can be used. Notably, the x in (17) represents the state T_i and $u = \{v, \Delta_x T_i, \Delta T_{o,i}\}$ the inputs, whereas x in the rest of the article denotes both the state and the inputs.

By using the target $\Delta_t T_i^{k+1}$ the mean function is implicitly encoded so that it preserves the state instead of going towards zero for inputs far away from the training points, a zero mean Gaussian process can thus be further assumed. For equal length segments the same transition function can be applied for every point in the entire grid, and as a consequence also trained on arbitrary points in the grid.

For the choice of kernel the interactions between the inputs and the state of (5) need to be included, while unnecessary terms should be avoided. This is accomplished using a so called additive kernel [11]. For each grid point, using the assumed structure given by (5) yields

$$k_i(x_i, x_i') = \underbrace{\sigma_1^2 \text{exp}\left(-\left(\frac{(v - v')^2}{2\ell_1^2} + \frac{(\Delta_x T_i - \Delta_x T_i')^2}{2\ell_2^2}\right)\right)}_{k_1(x_i, x_i')}$$
$$+ \underbrace{\sigma_2^2 \text{exp}\left(-\left(\frac{(\Delta T_{o,i} - \Delta T_{o,i}')^2}{2\ell_3^2}\right)\right)}_{k_2(x_i, x_i')}, \tag{20}$$

where $x_i = \{v, \Delta_x T_i, \Delta T_{o,i}\}$. This leaves five hyperparameters $\{\sigma_1, \sigma_2, \ell_1, \ell_2, \ell_3\}$ for learning the thermal dynamics of the entire grid. A schematic overview of the dynamic Gaussian process model can be seen in Figure 3.

B. Uncertainty propagation

For a multi step Gaussian process the output is not a Gaussian distribution in general, but can be approximated as such using moment matching [12] to handle uncertain inputs, when the inputs are given as a Gaussian distribution. The method is based on a Taylor expansion around the mean $\bar{f}_*$. To reduce the computational cost we consider only the first order terms in this paper. This leads to an unchanged posterior mean, and a correction term to the

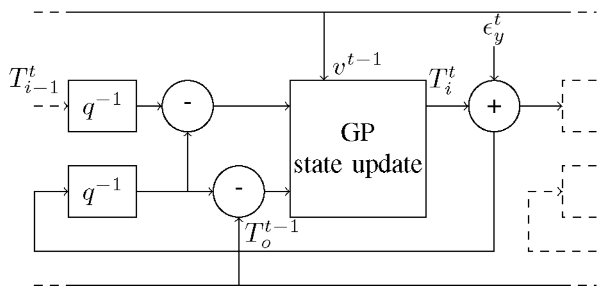

Fig. 3. Schematic overview of the dynamic Gaussian process model, where q^{-1} represent time lags.

posterior variance that becomes

$$\mathrm{cov}(f_*) = k_{**} - k_*^\top (K + \sigma_n^2 I)^{-1} k_* + \Delta_{\bar{f}}^\top \Sigma \Delta_{\bar{f}}, \quad (21)$$

where $\Delta_{\bar{f}}$ is the derivative of the training points with respect to the inputs

$$\Delta_{\bar{f}} = \frac{\delta \bar{f}_*}{\delta x_*}, \quad (22)$$

and is approximated from the noiseless posterior. For the squared exponential kernel used in this case, defined in equation (20), this can be calculated analytically, using

$$\alpha = (K + \sigma_n^2 I)^{-1} y, \qquad \frac{\delta \bar{f}_*}{\delta x_*} = \frac{\delta k_*}{\delta x_*} \alpha, \quad (23)$$

and defining $\tilde{X}_*^\top = [x_* - x_1, \ldots, x_* - x_N]^\top$ and the elementwise product $\circ$ yields the derivatives

$$\frac{\delta \bar{f}_*}{\delta v_*} = \ell_1^{-2} \tilde{v}_* \left(k_1(x_{i*}, X) \circ \alpha \right) \quad (24)$$

$$\frac{\delta \bar{f}_*}{\delta \Delta_x T_{i*}} = \ell_2^{-2} \Delta_x \tilde{T}_{i*} \left(k_1(x_{i*}, X) \circ \alpha \right) \quad (25)$$

$$\frac{\delta \bar{f}_*}{\delta \Delta T_{o,i*}} = \ell_3^{-2} \Delta \tilde{T}_{o,i*} \left(k_2(x_{i*}, X) \circ \alpha \right). \quad (26)$$

For proper propagation it needs to be taken into account that when using differences to get ΔT_i and $\Delta T_{o,i}$, the corresponding variances must be *added.* Using $\Sigma_i = \mathrm{diag}(\{\sigma_v, \sigma_{\Delta T_i}, \sigma_{\Delta T_{o,i}}\})$ results in the update

$$\sigma_v^{k+1} = \sigma_v^k \quad (27)$$

$$\sigma_{\Delta_x T_i}^{k+1} = \sigma_{T_{i-1}}^k + \sigma_{T_i}^k \quad (28)$$

$$\sigma_{\Delta T_{o,i}}^{k+1} = \sigma_{T_{o,i}}^k + \sigma_{T_i}^k \quad (29)$$

for each time step, where σ_v is the variance of the fluid velocity, $\sigma_{\Delta T_i}$ the variance of $\Delta_x T_i$, $\sigma_{\Delta T_{o,i}}$ the variance of ΔT_o, $\sigma_{T_{i-1}}$ the variance of the predicted temperature, and σ_{T_o} the variance of the (possibly) predicted outdoor temperature.

C. Sparse Gaussian process approximation

Gaussian process learning is $\mathcal{O}(n^3)$ where n is the number of data points. This is due to the inversion of K appearing in (13) and the determinant and inversion in (16). Clearly, it is preferable to limit the amount of data used for training. This is partly achieved by using the known physical relationships from section III, avoiding the exploration of an unnecessarily large covariate space.

To increase the computational performance of both training and prediction further, so called sparse Gaussian process methods can be used. In this paper a so called Subset of Data (SoD) method is used, where $m < n$ input points are chosen from a larger dataset of length n. This subset of points is chosen to maximize the differential entropy [13] from including an additional point, so that the information gain for each added point is maximized. The entropy is given by

$$H(x) = \frac{1}{2}\log|K| + \frac{d}{2}\log(2\pi e), \quad (30)$$

where d is the dimension of the input. While the determinant of K is a relatively costly operation, the size of the kernel is initially small and the training can be performed with reasonable speed. The algorithm depends on that the hyperparameters are already known, this is achieved by first training the non-sparse Gaussian process on n_{pre} measurement points, picked according to an uniform random distribution.

V. Results

The Gaussian process model is implemented in the Julia language [14], with the Optim package [15] and the quasi Newton L-BFGS solver for finding the hyperparameters. Forward simulation, i.e calculating each grid point, is a relatively computationally cheap operation since the costly matrix inversions of (13-14) can be done once and for all. The forward simulation algorithm is summarized in Algorithm 1.

Algorithm 1: Forward simulation of the GPSSM

```
Result: Vector T
T = T_init;
for k in k_steps do
    for i in n_x do
        Calculate posterior distribution of GP f(x_i^k);
    end
    Update states T^k = T^{k-1} + f̄(x^k);
    Update inputs x^{k+1};
    Update uncertainties Σ^k;
end
```

For training the model, the well validated pipe model from the Modelica Standard Library [16] has been used as a reference. For training and validation purposes it is advantageous that the states along the pipe length are accessible, so that it can seamlessly be used for training the Gaussian process surrogate model. The input signals are increasing frequency sinusoidals, known as chirps, that empirically provided good model performance. Amplitudes and frequencies were chosen to reflect typical variations for temperature and flow for a district heating pipe. For the validation a pipe with a length of $500m$ discretized with 50

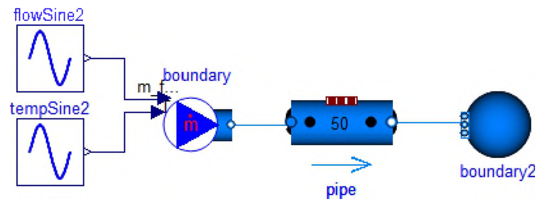

Fig. 4. Graphical overview of the Modelica model with one long district heating pipe and sinusoidally varying inlet flow rate and temperature.

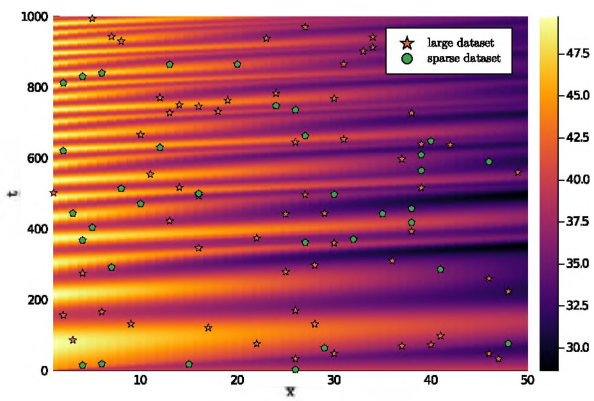

Fig. 5. Heatmap of the training dataset, where the colors represent the temperature. The large dataset (stars) are the points randomly picked from a uniform distribution. Points chosen for the sparse subset of data represented as circles.

segments is used. A graphical view of the Modelica model can be seen in Figure 4.

The model is first trained with $n = 100$ random points from a uniform distribution to get reasonable values for the hyperparameters. A sparse approximation is then found from a larger dataset of $n = 1000$ points, reduced to the final sparse approximation of $m = 25$ points, where the hyper-parameters are once again optimized using L-BFGS. The dataset visualized as a heatmap over the varying temperature in time and space is shown in Figure 5, with the n and m points overlayed.

For the validation dataset, the temperature and flow rate entering the pipe are sinusoidals with amplitudes $45 \pm 4\,°\mathrm{C}, 10 \pm 3\,\mathrm{kg\,s^{-1}}$ and frequencies $40 \times 10^{-5}\,\mathrm{Hz}, 15 \times 10^{-5}\,\mathrm{Hz}$ respectively, reflecting reasonable variations of district heating temperature and flow. Validation results show a good correspondence between the simulated pipe output temperature and the Modelica model (Figure 6), as well as for the spatial distribution (Figure 7) along the pipe length. The root mean square error (RMSE) is calculated as

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}(T_{GP} - T_{val})^2}, \tag{31}$$

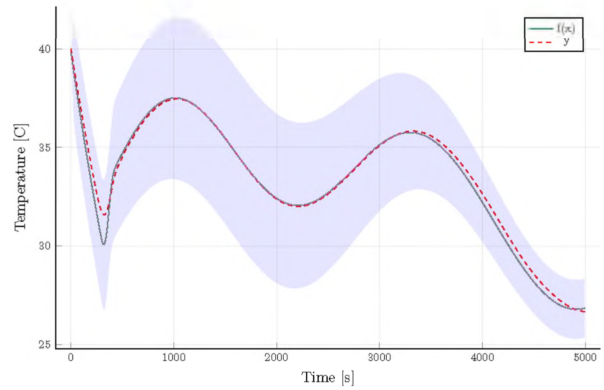

Fig. 6. Water temperature of pipe exit at 500m, simulation with 50 segments, Gaussian process approximation (green) and Modelica simulation (red, dashed). Two standard deviations plotted as shaded area.

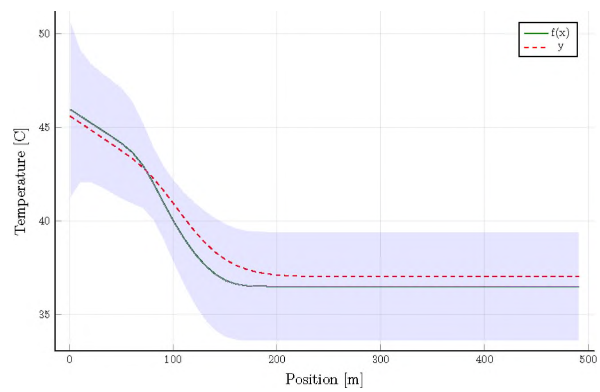

Fig. 7. Water temperature along the pipe length at t=250s, simulation with 50 segments, Gaussian process approximation (green) and Modelica simulation (red, dashed). Two standard deviations plotted as shaded area.

where n is the number of measurement points, T_{GP} the temperature output of the Gaussian process model and T_{val} the temperature output of the Modelica model. For the validation dataset the RMSE ≈ 0.36, which is roughly comparable to the RMSE between the Modelica model and an experimental setup [7]. A heatmap of the error in Figure 8 shows that for the simulated example the error is mostly within $\pm 1\,°\mathrm{C}$ for the time range of 5000s.

Using 50 segments, the simulation (prediction) runs in $\approx$ 5000 time real-time speed on a standard 4 core Intel 8th gen i5 laptop. As expected, the computational cost scales linearly with the number of segments.

For validation of the uncertainty propagation a Monte Carlo simulation was chosen as reference, where T_o, T_{in} and v are sampled from a Gaussian distribution. The variances were chosen as $\Sigma = \mathrm{diag}(\{5.0, 5.0, 1.0\})$, reflecting variances that typically occur from temperature and flow rate predictions. The results show that the propagated uncertainty matches well with the Monte Carlo simulations, as can be seen in Figure 9.

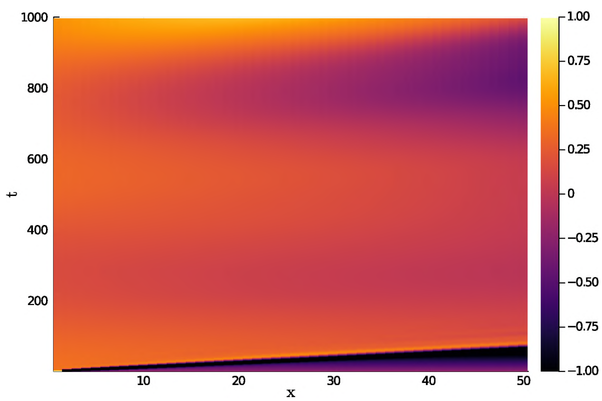

Fig. 8. Heatmap of the error for the validation dataset, with 50 segments and 1000 timesteps of $\Delta t = 5s$.

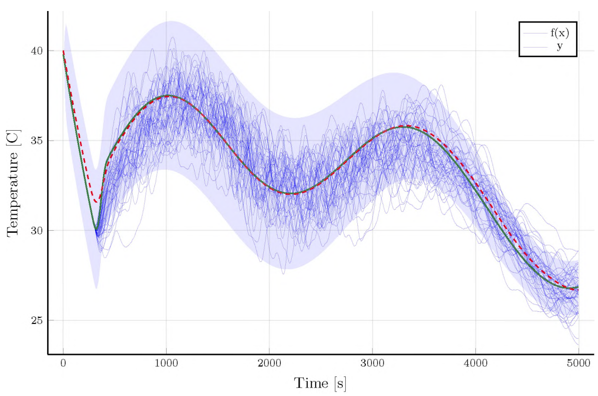

Fig. 9. Monte Carlo simulation of the water temperature at the pipe exit at 500m, simulation with 50 segments. Temperature trajectories from the sampled inputs are shown in blue, Gaussian process approximation (green) and Modelica simulation (red. dashed). Two standard deviations plotted as shaded area.

VI. Conclusions

In this study a method for modelling of a thermal grid on the basis of Gaussian processes is suggested. The results show that the method can approximate a dynamic simulation model of a district heating pipe with reasonable accuracy, using domain knowledge of the physical model and numerical methods. The results indicate that it is feasible to extend the use of the Gaussian process state space model to larger parts of the grid.

The result is a Gaussian process state space model where the uncertainty of the inputs is propagated in a way that can be employed e.g for automatic control at optimization methods. The actual implementations of these methods are not considered in the article, but are seen as interesting areas for future research.

For future work the approach needs to be scaled up to a larger grid. Here, the probabilistic view of the simulation can provide an alternative approach to e.g simulation related topological optimization of the network to further enhance the computational performance. The current implementation also does not consider bidirectional flow, that is present in some district heating grids. For the Julia implementation in the paper the computational performance has not been the main target, and optimization of the code would further extend the usability of the method.

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